

QUANT RUSH · WORKSHOP 1

PICKS & SHOVELS I

Python, pandas, and your first submission.

TUE OCT 20

ZERO TO A BASELINE

Lithograph: N. Currier, “The Independent Gold Hunter on His Way to California,” c. 1849 · Missouri History Museum · Public domain

TODAY

Zero to a submitted baseline

No experience assumed. Four steps, one hour, and everyone leaves with a score on the leaderboard.

1
OPEN
a notebook on Kaggle

2
LOAD
the data with pandas

3
PREDICT
with a one-line rule

4
SUBMIT
a checked file

KAGGLE NOTEBOOKS

A notebook, cell by cell

START

On the competition page: Code tab → New Notebook. The competition files are already attached.

RUN

Shift + Enter runs a cell. Code cells run Python; text cells explain.

SAVE

Save Version runs everything from the top and keeps the files your notebook writes.

Free, in the browser, nothing to install.

Values, lists, functions

```
hours = 12
channels = ["ret", "vol", "range", "taker"]

def column_names(channel):
    return [f"{channel}_{k:02d}" for k in range(hours)]

column_names("ret")
→ ['ret_00', 'ret_01', …, 'ret_11']
```

A VALUE

gets a name with =.

A LIST

holds several values in [ ].

A FUNCTION

is a reusable step. def names it; return hands back the result.

PANDAS

Tables in Python

```
import pandas as pd

train = pd.read_parquet(DATA / "train.parquet")
train.shape
→ (1181414, 75)

train.head()
→ the first 5 rows
```

DATAFRAME

A whole table: rows and named columns.

SERIES

One column of it.

PARQUET

A fast file format. The CSV files hold the same data.

FIVE LINES TO KNOW

Pick columns, pick rows

Code	What you get
`train["ret_00"]`	one column (a Series)
`train[["id", "ret_00"]]`	several columns (a DataFrame)
`train[train["era"] == 0]`	only the rows of era 0
`train.groupby("era").size()`	how many assets each era has
`train.describe()`	count, mean, and spread of every column

One pair of brackets picks a column. A condition inside the brackets picks rows.

THE DATA

One row: one asset, one era

`id, era`

The row’s id, and the moment it belongs to. Train eras are numbered in time order.

`72 inputs`

Return, volume, range, and buying pressure for each of the last 12 hours, plus the whole market.

`target`

How the asset did over the next 4 hours, as a rank within its era from −0.5 to +0.5.

The test set has the same columns without target. That is what you predict: one number per row.

A FIRST IDEA

Short-term reversal

What jumped in the last hour tends to give a little of it back. What dropped tends to recover a little.

So rank the assets by the opposite of their last hour's return. Higher prediction means you expect a higher rank.

```
prediction = -ret_00
```

One line. No machine learning yet: that part is up to you.

CHECK IT FIRST

Score it on train before you submit

Four rules, scored with the competition metric on the last 1,080 training eras:

Rule	Score
-ret_00 (reversal)	0.0576
+ret_00 (the opposite)	-0.0576
-ret_01 (one hour older)	0.0299
random guess	0.0006

5 A DAY

Five submissions a day per team. Train has the answers: test ideas there.

RECENT ERAS

Check on the latest training eras: the test set comes after them in time.

THE SUBMISSION FILE

Two columns, one row per test row

```
submission = pd.DataFrame({
    "id": test["id"],
    "prediction": -test["ret_00"],
})
submission.to_csv("submission.csv", index=False)
```

SUBMISSION.CSV

```
id,prediction
86eaa3e2e244,0.47
ca9517ab9d09,-1.05
…
```

Kaggle rejects a missing id, a repeated id, or a prediction that is not a number.

THE CLICKS

Submitting on Kaggle

1

SAVE VERSION

Runs the notebook top to bottom and keeps submission.csv.

2

SUBMIT

On the competition page, open Submit Predictions.

3

PICK THE FILE

Your notebook's submission.csv, or a file you downloaded.

4

READ THE SCORE

Your public score shows on the leaderboard.

Before the deadline, pick 2 final submissions. If you pick none, Kaggle uses your best public scores.

YOUR FIRST SCORE

About 0.04. Now beat it.

The reversal rule is the Benchmark line on the leaderboard: 0.039 on the public part of the test set.

The public score covers part of the test set. The private score, revealed after the deadline, decides the winners.

TRY IT THIS WEEK

Beat the benchmark

```
recent["prediction"] = -recent["ret_00"]  # change this line
print(era_score(recent))
```

1 EXPLORE

Score other single columns on recent eras, with and without a minus sign.

2 PICK

Choose your best rule and write a checked submission.csv.

3 SUBMIT

Get on the leaderboard, then share what you tried on Discord.

RECAP

Your tools are ready

- A notebook runs Python cell by cell; pandas holds the data.
- Pick columns with ["col"], rows with a condition, eras with groupby.
- Only the order within each era is scored.
- Check locally, then submit a checked file.

Next · Tue Oct 27

READING THE VEIN

Workshop 2: exploring the data and understanding the metric.

QUESTIONS?

Ask now, or any time on Discord: discord.gg/MUPdvSAV4M